This drawing standard is approved for use by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for approved use shall be made from this document.

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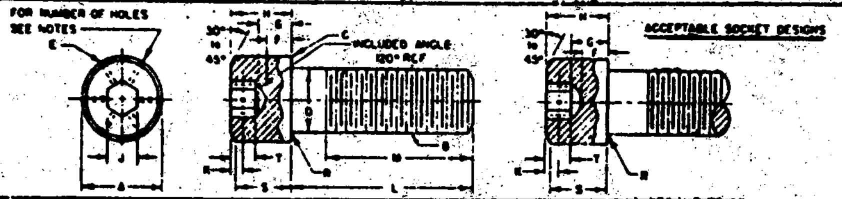

NOMINAL SIZE		#4 (.112)	#6 (.138)	#8 (.164)	#10 (.190)	1/4
S- THREADS PER INCH		40UNC-3A	32UNC-3A	32UNC-3A	24UNC-3A	20UNC-3A
D- BODY DIAMETER	MAX MIN	.1120 .1075	.1380 .1329	.1640 .1585	.1900 .1840	.2500 .2435
A- HEAD DIAMETER	MAX MIN	.183 .176	.226 .218	.270 .262	.312 .303	.375 .365
H- HEAD HEIGHT	MAX MIN	.112 .108	.138 .134	.164 .159	.190 .185	.250 .244
F- [illegible] DEPTH	MIN	.009	.013	.011	.015	[illegible]
S- HEAD SIDE HEIGHT	MIN	.101	.124	.148	.171	[illegible]
J- SOCKET WIDTH ACROSS FLATS	MAX MIN	.0952 .0937	.1111 .1094	.1426 .1406	.1587 .1562	[illegible] [illegible]
T- KEY ENGAGEMENT	MIN	.051	[illegible]	.077	[illegible]	[illegible]
G- WALL THICKNESS	MIN	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible] RADIUS OR CHAMFER	MAX	.005	.005	.005	.005	.005
R- RADIUS	MAX MIN	.009 .005	.010 .006	.012 .007	.014 .009	.014 [illegible]
K- HOLE LOCATION	MAX MIN	.040 .026	.050 .035	.060 .040	.065 .045	[illegible] [illegible]
E- HOLE DIAMETER	MAX MIN	.039 .033	.039 .033	.050 .044	.050 [illegible]	.050 .044
M- MINIMUM THREAD LENGTH		.750	.750	.875	.875	1.000
RECOMMENDED WIRE DIAMETER	MAX MIN	.028 .020	.028 .020	.040 .032	.040 .032	.040 .032
*DRILL HOLE ALIGNMENT PLUG		.025	.025	.030	.030	.030
**MINIMUM BREAKING STRENGTH		980 LB	1,400 LB	2,170 LB	2,710 LB	[illegible] LB

L- LENGTH	PART NO.		PART NO.		PART NO.		PART NO.		PART NO.	
1/4 3/8 1/2 5/8 3/4 7/8 1 1-1/4 1-1/2 1-3/4 2 2-1/4 2-1/2 2-3/4	MS 24676 -1 -2 -3 -4 MS 24676 -5		MS 24676 -6 -7 -8 -9 -10 -11 MS 24676 -12		MS 24676 -13 -14 -15 -16 -17 -18 -19 -20 MS 24676 -21		MS 24676 -22 ↕ MS 24676 -34		MS 24676 -3[illegible] ↕ MS 24676 -[illegible]	

Ⓑ FOR CHANGES SEE SHEETS 1,2,3,4

Ⓑ CANCELLED

AFTER 7 JULY 1975 USE MS 24677

SEE SHEET 4

APPROVED 7 APRIL 1960 REVISED Ⓐ 27 JAN 1961 Ⓑ 7 JULY 1975

P.A. [illegible]

Other Cust
C/D
ARS

PROCUREMENT SPECIFICATION
[illegible]

TITLE
SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED
ALLOY STEEL, [illegible]ATED

SUPERSEDES

MILITARY STANDARD

MS 24676

SHEET 1 OF 4

FED SUP CLASS
[illegible]

This military standard is approved by the Department of Defense and is mandatory for use by all Departments and Agencies. [illegible]

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; [illegible]

NOMINAL SIZE		5/16	3/8	7/16	1/2	5/8
THREADS PER INCH		18UNC-3A	16UNC-3A	14UNC-3A	13UNC-3A	11UNC-3A
D- BODY DIAMETER*	MAX	.3125	.3750	.4375	.5000	.6250
	MIN	.3053	.3678	.4294	.4919	.6163
A- HEAD DIAMETER	MAX	.468	.562	.656	.750	.937
	MIN	.457	.550	.642	.735	.921
H- HEAD HEIGHT	MAX	.312	.375	.437	.500	.625
	MIN	.306	.369	.430	.492	.615
HOLE DEPTH	MIN	.027	.029	.031	.351	[illegible]
S- HEAD SIDE HEIGHT	MIN	.221	.337	.334	.436	[illegible]
J- SOCKET WIDTH ACROSS FLATS	MAX	.2530	.3155	.3780	.3790	.5050
	MIN	.2520	.3125	.3750	.3750	.5020
T- KEY ENGAGEMENT	MIN	.151	.182	.213	.245	[illegible]
G- WALL THICKNESS	MIN	.102	.130	.140	.153	[illegible]
RADIUS OR CHAMFER	MAX	.008	.008	.010	.010	.010
R- RADIUS	MAX	.017	.020	.023	.025	.032
	MIN	.012	.015	.018	.020	.024
E- HOLE LOCATION	MAX	.104	.123	.141	.160	.192
	MIN	.084	.133	.121	.140	.172
F- HOLE DIAMETER	MAX	.050	.067	.067	.067	.067
	MIN	.044	.061	.061	.061	.061
M- MINIMUM THREAD LENGTH		1.125	1.250	1.375	1.500	1.750
RECOMMENDED WIRE DIAMETER	MAX	.040	.057	.057	.057	.057
	MIN	.032	.055	.055	.055	.055
*DRILL HOLE ALIGNMENT PLUG		.030				
**MINIMUM BREAKING STRENGTH		8,190 LB	12,100 LB	16,700 LB	22,300 LB	[illegible] LB

L- LENGTH	5/16 PART NO.		3/8 PART NO.		7/16 PART NO.		1/2 PART NO.		5/8 PART NO.	
3/8	MS 24676 -46									
1/2	-47		MS 24676 -62		MS 24676 -80		MS24676 -96			
5/8	-48		-63		-81		-97			
3/4	-49		-64		-82		-98			
7/8	-50		-65		-83		-99			
1	-51		-66		-84		-100		MS24676-116	
1-1/4	-52		-67		-85		-101		-117	
1-1/2	-53		-68		-86		-102		-118	
1-3/4	-54		-69		-87		-103		-119	
2	-55		-70		-88		-104		-120	
2-1/4	-56		-71		-89		-105		-121	
2-1/2	-57		-72		-90		-106		-122	
2-3/4	-58		-73		-91		-107		-123	
3	-59		-74		-92		-108		-124	
3-1/4	-60		-75		-93		-109		-125	
3-1/2	MS 24676 -61		-76		-94		-110		-126	
4			-77		MS 24676 -95		-111		-127	
4-1/2			-78				-112		-128	
5			MS24676-79				-113		-129	
5-1/2							-114		-130	
6							MS24676-115		MS24676-131	

Ⓑ CANCELLED

AFTER 7 JULY 1975 USE MS 24677

SEE SHEET 4

FSC 5305

APPROVED 7 APRIL 1960 REVISED Ⓑ FOR CHANGES SEE SHEETS 1, 2, 3, 4

P.A. [illegible]

Other Cust
ORD
AMD

TITLE

SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED
ALLOY STEEL, UNCOATED
(UNC-3A)

PROCUREMENT SPECIFICATION
FF-S-86

SUPERSEDES

MILITARY STANDARD

MS 24676

SHEET 2

[illegible]

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

NOMINAL SIZE		3/4	7/8	1		
[illegible]		10UNC-3A	9UNC-3A	8UNC-3A		
D- BODY DIAMETER	MAX MIN	.7500 .7406	.8750 .8647	1.0000 .9886		
A- HEAD DIAMETER	MAX MIN	1.125 1.107	1.312 1.291	1.500 1.479		
[illegible]	MAX MIN	.750 .740	.875 .865	1.000 .990		
[illegible]	MIN	[illegible]	[illegible]	[illegible]		
[illegible] HEAD SIDE HEIGHT	MIN	[illegible]		[illegible]		
[illegible] SOCKET WIDTH ACROSS FLATS	MAX MIN	.6312 .6259	.7575 .7500	.7575 .7500		
[illegible]	MIN	[illegible]	[illegible]	[illegible]		
[illegible]	MIN	.245	[illegible]	.320		
[illegible] OR CHAMFER	MAX	.013	[illegible]	[illegible]		
[illegible]	MAX MIN	.039 .030	[illegible]	.050 .040		
[illegible]	MAX MIN	.235 .215	.273 [illegible]	.310 .290		
[illegible]	MAX MIN	.097 .052	[illegible]	[illegible] .081		
[illegible]		2.000	2.250	[illegible]		
[illegible]	MAX MIN	.090 [illegible]	[illegible]	[illegible]		
[illegible]		[illegible] LB	[illegible] LB	[illegible] LB		

L- LENGTH	PART NO.		PART NO.		PART NO.	
1-1/4	MS24676-132					
1-1/2	-133					
1-3/4	-134					
2	-135		MS24676-147		MS24676-159	
2-1/4	-136		-148		-160	
2-1/2	-137		-149		-161	
2-3/4	-138		-150		-162	
3	-139		-151		-163	
3-1/4	-140		-152		-164	
3-1/2	-141		-153		-165	
4	-142		-154		-166	
4-1/2	-143		-155		-167	
5	-144		-156		-168	
5-1/2	-145		-157		-169	
6	MS24676-146		MS24676-158		MS24676-170	

Ⓑ CANCELLED

AFTER 7 JULY 1978 USE MS 24677

SEE SHEET 4

APPROVED 7 APRIL 1960 REVISED Ⓑ FOR CHANGES SEE SHEETS 1, 2, 3, 4

FSC 5305

SCREW, CAP-SOCKET HEAD, HEXAGON, DRILLED, [illegible] (UNC-3A)

MILITARY STANDARD

MS24676

DD FORM 672-1

This military standard is approved by the Department of Defense and is mandatory for use by all activities. Selection for all new engineering and design applications and for repurchases shall be made from this document.

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*Alignment Plug shall pass completely through the head without deflection.

**Based on 160,000 PSI minimum tensile strength. Breaking strengths are determined by the stress areas indicated in Screw-Thread Standards for Federal Services Handbook H-28.

MATERIAL: Steel, Alloy: Federal Standard 66.

HARDNESS: Rockwell, C36-43

PROTECTIVE COATING: Uncoated.

THREADS: The threads shall be in accordance with Specification MIL-S-7742

THREAD LENGTH-A: Screws which have a body length less than the minimum thread length shall be threaded within two thread pitches of the head.

LENGTH TOLERANCE:

LENGTH	DIAMETER	TOLERANCE
1/4 thru 1.000	#4 thru 1.000	±1/64
1-1/4 thru 2.000	#4 thru 1.000	+1/32 -1/64
2-1/4 thru 6.000	#4 thru 1.000	±1/32

DRILLED HOLE DATA:

(a) Screw sizes No. 4 and No. 6 shall have two (2) drilled holes spaced 180° and size No. 8 and above shall have six (6) drilled holes spaced 60°
(b) Edges of the holes on the outside of the head shall be chamfered 45° by .010 to .020 deep.
(c) On screw sizes below No. 8 the drilled hole shall be centered as close as practicable on the flat of the hexagon socket.
(d) On screw sizes No. 8 and above the drilled holes shall be within the flats of the hexagon socket and shall not break through the corners of the hexagon socket.

HEAD: Plain or knurled heads are optional.

DIMENSIONS: In inches unless otherwise specified.

NOTES:

(a) This Standard takes precedence over documents referenced herein.
(b) Reference documents shall be of the issue in effect on date of invitations for bids.

PART NUMBER = (MS Number) - (Screw dash number)
EXAMPLE = MS 24676 -1 is the part number for a screw #4-40NC-3A by 1/4 long Alloy Steel uncoated.

Ⓑ CANCELLED

AFTER 7 JULY 1976 USE MS 24677

INTERCHANGEABILITY RELATIONSHIP – MS 24677 SCREWS CAN REPLACE THE CANCELLED SCREWS OF THE SAME DASH NUMBERS FOR MOST APPLICATIONS, EXCEPT THOSE IN WHICH CADMIUM PLATED STEEL SCREWS ARE UNACCEPTABLE DUE TO TEMPERATURE, CORROSION, OR TOXICITY CONSIDERATIONS. THE CANCELLED MS 24676 SCREWS CANNOT ALWAYS REPLACE MS 24677 SCREWS AND SHOULD BE USED UNTIL EXISTING STOCK IS DEPLETED. USE ONLY SUPERSEDING MS SCREWS OF THE SAME DASH NUMBERS FOR DESIGN AND REPLACEMENT.

FED. SUP. CLASS
5305

APP 7 APRIL 1960 REVISED Ⓑ FOR CHANGES SEE SHEETS 1, 2, 3, 4

P.A. ...
Other Cust
ORD
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TITLE
SCREW, CAP - SOCKET HEAD, HEXAGON, DRILLED
ALLOY STEEL, UNCOATED
(UNC-3A)

PROCUREMENT SPECIFICATION
FF-S-86

SUPERSEDES

MILITARY STANDARD

MS 24676

DD FORM 672-1